

The University of Texas at Austin
IN-CLASS WORK 5

M339G Predictive Analytics

October 04, 2026

LOGISTIC REGRESSION.

Problem 5.1. You run a logistic regression to predict the binary response Y , indicating whether a credit card holder will default on a payment in the next six months, from a single numerical explanatory variable X (number of late payments in the past year).

You are given the following estimated coefficients:

- $\hat{\beta}_0 = -3.5$
- $\hat{\beta}_1 = 0.8$

Calculate the fitted **odds** for a card holder with three late payments. Then, convert the odds to the corresponding **probability of success**.

Problem 5.2. An insurer uses logistic regression to model the probability that a policyholder files a claim in the next year, using a single numerical explanatory variable X (age of the insured vehicle, in years). The estimated slope coefficient is positive.

Determine whether each of the following statements is **TRUE** or **FALSE**. Justify each answer in one or two sentences.

1. The fitted probability of filing a claim increases as the age of the vehicle increases.
2. Each additional year of vehicle age increases the fitted probability of filing a claim by the same amount.
3. Each additional year of vehicle age multiplies the fitted odds of filing a claim by the same factor.
4. The fitted probabilities always lie strictly between 0 and 1, regardless of the value of X .
5. Ordinary least squares regression of Y on X would be an equally appropriate model for these data.

Problem 5.3. You are using logistic regression to predict the probability that a life insurance policy will lapse in the next year. Your predictor is a categorical random variable indicating the *Payment* frequency: *Monthly*, *Quarterly*, *Annual*. The *Monthly* category is understood as the *baseline*. You obtain the following summary of coefficients:

Intercept	−1.842
PaymentQuarterly	−0.613
PaymentAnnual	−1.307

What is the fitted probability that a policy with *Quarterly* payments will lapse?

Problem 5.4. You are using logistic regression to predict the probability of a particular class of driver having a claim in the next insurance period. Your predictors are two categorical random variables indicating the

- *Region* in which the driver lives: *North*, *South*, *East*, *West* with *North* as *baseline*.
- *Use* of the vehicle: *Personal*, *Commute*, *Business* with *Personal* as the *baseline*.

You obtain the following summary of coefficients:

Intercept	-1.732
RegionSouth	0.128
RegionEast	-0.064
RegionWest	0.211
UseCommute	0.342
UseBusiness	0.587

What is the fitted probability that a driver from the *West* region who uses the vehicle for *Business* will have a claim?

Problem 5.5. In a study, 200 customers were asked to choose one of three insurance plans (Basic, Standard, Premium). The customers were organized into four income categories (Low, Lower-middle, Upper-middle, High).

A logistic regression was fitted to the customers' responses to predict their preferred plan with income category Low and the Basic plan as reference categories.

For income category Low the log-odds for preference of the Standard plan and the Premium plan were -0.412 and -1.127 , respectively.

Calculate the modeled probability of someone from income category Low preferring the Premium plan.

Problem 5.6. A statistician uses a logistic model to predict the probability of success, π , of a binomial random variable.

You are given the following information:

- There is one predictor random variable, X , and an intercept in the model.
- The estimates of π at $x = 2$ and $x = 5$ are 0.57444 and 0.92757, respectively.

Calculate the estimated intercept coefficient, b_0 , and the slope coefficient, b_1 , in the logistic model that produced the above probability estimates.

